1 Fermat's and Euler's Theorems, Inverses and Primitive Roots

Public-key cryptography computes powers a^b mod n with exponents hundreds of digits long, and it needs inverses modulo n. Square-and-multiply (§**??**) makes such powers feasible, but the two theorems of this section do more: they say that the powers of a modulo n repeat with a known period, so a huge exponent can be replaced by a small one, and they give a formula for the inverse. The same periodicity, studied element by element, leads to the order of an element and to primitive roots, which Diffie–Hellman and ElGamal are built on.

1.1 Fermat's little theorem and exponent reduction ★

★ **In the papers:** 2024–25 Q5(ii) (part of 3 marks): 100^{95} mod 9, 658^{17} mod 17, 145^{102} mod 101.

First version. If p is prime and a is an integer not divisible by p, then $a^{p-1} \equiv 1 \pmod{p}$. Both conditions must be checked before the theorem is used: for a composite modulus or for $p \mid a$ it says nothing.

Why it is true. Multiply each of $1, 2, \ldots, p-1$ by a and reduce modulo p. No product is 0, because p divides neither factor, and no two products are equal, because $ai \equiv aj$ would mean $p \mid a(i-j)$ and hence $p \mid (i-j)$, i.e. $i = j$. The $p-1$ products are therefore just $1, \ldots, p-1$ in a different order; for $p = 7$, $a = 3$ they are $3, 6, 2, 5, 1, 4$. Multiplying all of them together gives $a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$, and since $(p-1)!$ is co-prime to p it can be cancelled, leaving $a^{p-1} \equiv 1$.

Second version. If p is prime, then $a^p \equiv a \pmod{p}$ for every integer a. This removes the condition on a. If $p \nmid a$, it is the first version multiplied by a; if $p \mid a$, both sides are $\equiv 0$.

p, a	Conditions	Working	Result
5, 2	prime, $5 \nmid 2$	$2^4 = 16 = 3 \cdot 5 + 1$	$\equiv 1$ ✓
13, 11	prime, $13 \nmid 11$	$11 \equiv -2$, $(-2)^{12} = \left((-2)^4\right)^3 = 16^3 \equiv 3^3 = 27 \equiv 1$	$\equiv 1$ ✓
6, 2	**6 not prime**	$2^5 = 32 = 5 \cdot 6 + 2$	$\equiv 2 \neq 1$: theorem silent
11, 5 (homework)	prime, $11 \nmid 5$	$5^2 \equiv 3$, $5^4 \equiv 9$, $5^8 \equiv 81 \equiv 4$, $5^{10} = 5^8 \cdot 5^2 \equiv 4 \cdot 3 = 12$	$\equiv 1$ ✓

Exponent reduction. Since $a^{p-1} \equiv 1$, the powers of a modulo p repeat with period dividing $p-1$. Writing the exponent as $b = q(p-1) + r$ with $r = b \bmod (p-1)$ gives

$$a^b = \left(a^{p-1}\right)^q a^r \equiv 1^q \cdot a^r = a^{\,b \bmod (p-1)} \pmod{p} \qquad (p \text{ prime},\ p \nmid a).$$

The slides' examples follow directly: 6^{10} mod 11 = 1 because the exponent is exactly $p-1$, and 3^{12} mod 11 = $3^{11} \cdot 3 \equiv 3 \cdot 3 = 9$ by the second version (equivalently 12 mod 10 = 2 and $3^2 = 9$). In practice the base is reduced modulo p first, then the exponent modulo $p-1$, and the small power that remains is computed by repeated squaring.

Trap: what to reduce, and by what

The **base** is reduced modulo p; the **exponent** is reduced modulo $p-1$, never modulo p. And the modulus must be prime: in 100^{95} mod 9 the modulus $9 = 3^2$ is composite, so Fermat does not apply. There the base reduction alone finishes the job ($100 \equiv 1 \pmod{9}$), or Euler's theorem with $\phi(9) = 6$ can be used.

WORKED EXAMPLE : exponent reduction by Fermat (Forouzan exercise)

Q. Using Fermat's little theorem, find (a) 5^{15} mod 13 (b) 15^{18} mod 17 (c) 456^{17} mod 17.
A.

	Conditions	Base	Exponent	Result
(a)	13 prime, $13 \nmid 5$	5	$15 = 1 \cdot 12 + 3 \to 3$	$5^3 = 125 = 9 \cdot 13 + 8 \Rightarrow$ **8**
(b)	17 prime, $17 \nmid 15$	$15 \equiv -2$	$18 = 1 \cdot 16 + 2 \to 2$	$(-2)^2 =$ **4**
(c)	17 prime	$456 = 26 \cdot 17 + 14 \to 14$	second version: $a^{17} \equiv a$	**14**

Check (by script): 5^{15} mod 13 = 8, 15^{18} mod 17 = 4, 456^{17} mod 17 = 14.

Fermat as a primality test. Read backwards, the theorem gives a test: if $a^{n-1} \not\equiv 1 \pmod{n}$ for some a with $\gcd(a, n) = 1$, then n is certainly composite (e.g. 2^{14} mod 15 = 4). The converse fails: $2^{340} \equiv 1 \pmod{341}$ although $341 = 11 \cdot 31$. The Fermat test, Carmichael numbers and Miller–Rabin are covered in §**??**.

1.2 Euler's theorem ★

▸ **Likely this year:** Exponent reduction modulo $\phi(n)$ for a composite n, and "why does RSA decryption work?" (Euler is its basis).

Fermat's theorem needs a prime modulus, but RSA works modulo $n = pq$. Euler's theorem is the generalisation

to any modulus. **First version:** if $\gcd(a,n)=1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$. When $n=p$ is prime, $\phi(p)=p-1$ and this is exactly Fermat's theorem.

Why it is true. The proof is the same permutation argument, run over $\mathbb{Z}_n^*$ (§**??**) instead of $\{1,\dots,p-1\}$. Multiplying the $\phi(n)$ elements of $\mathbb{Z}_n^*$ by a permutes them, because a is invertible; for $n=10$, $a=3$ the set $\{1,3,7,9\}$ becomes $\{3,9,1,7\}$. Taking the product of all elements on both sides gives $a^{\phi(n)} \cdot \Pi \equiv \Pi$, and the product Π is invertible, so $a^{\phi(n)} \equiv 1$.

The co-prime condition is genuinely needed. If $\gcd(a,n)=d>1$, every power of a is a multiple of d, and so is every number $\equiv a^k \pmod{n}$; such a number can never be $\equiv 1$. The lecture homework $11^{20} \bmod 33$ shows this: the powers of 11 alternate $11, 22, 11, 22, \dots$, so $11^{20} \equiv 22 \neq 1$ although $\phi(33)=20$.

a, n	gcd	Working	Result
3, 10	1	$\phi(10)=4$; $3^4=81=8\cdot10+1$	$\equiv 1$ ✓
2, 10	**2**	$2^4=16\equiv6$	$\neq 1$: theorem silent
10, 11	1	$\phi(11)=10$; $10\equiv-1$, $(-1)^{10}=1$	$\equiv 1$ ✓
6, 35	1	$\phi(35)=24$; slide example $6^{24} \bmod 35$	$\equiv 1$ ✓
11, 33 (homework)	**11**	$\phi(33)=20$; powers alternate $11, 22$	$11^{20}\equiv22\neq1$

Exponent reduction. Exactly as with Fermat, $\gcd(a,n)=1$ and $b=q\phi(n)+r$ give $a^b \equiv a^{b \bmod \phi(n)} \pmod{n}$. The work is in computing $\phi(n)$, which needs the factorisation of n.

Second version. For $n=pq$ (and more generally any squarefree n), $a^{k\phi(n)+1} \equiv a \pmod{n}$ for every integer a and every $k \geq 0$. This removes the co-prime condition, which RSA needs because a message M might, by bad luck, share the factor p with n. The proof works one prime at a time. Modulo p: if $p \nmid a$, then $a^{k\phi(n)} = \left(a^{p-1}\right)^{k(q-1)} \equiv 1$ by Fermat, so $a^{k\phi(n)+1} \equiv a$; if $p \mid a$, both sides are $\equiv 0$. The same holds modulo q. Since the distinct primes p and q both divide $a^{k\phi(n)+1}-a$, so does pq. For example $11^{21} = 11^{\phi(33)+1} \equiv 11 \pmod{33}$ although $\gcd(11,33)\neq1$. The argument breaks when a prime appears squared: for $n=4$, $a=2$, $k=1$ the claim would be $2^3 \equiv 2$, but $2^3=8\equiv0 \pmod{4}$.

Why RSA decryption works. RSA picks e and d with $ed \equiv 1 \pmod{\phi(n)}$, i.e. $ed=k\phi(n)+1$. Decrypting $C=M^e$ gives $C^d = M^{ed} = M^{k\phi(n)+1} \equiv M \pmod{n}$ by the second version, for every message $M<n$. In the example $n=187$, $e=7$, $d=23$: $ed=161=1\cdot160+1$, and indeed $88^7 \bmod 187 = 11$ and $11^{23} \bmod 187 = 88$.

★ MUST KNOW

	First version	Second version (no co-prime condition)
Fermat (p prime)	$a^{p-1}\equiv1 \pmod{p}$ if $p\nmid a$	$a^p \equiv a \pmod{p}$
Euler	$a^{\phi(n)}\equiv1 \pmod{n}$ if $\gcd(a,n)=1$	$a^{k\phi(n)+1}\equiv a \pmod{n}$ ($n=pq$, squarefree)
Reduction	$a^b \equiv a^{b \bmod (p-1)} \pmod{p}$ and	$a^b \equiv a^{b \bmod \phi(n)} \pmod{n}$ (co-prime a)
Inverse	$a^{-1} \equiv a^{p-2} \pmod{p}$ and	$a^{-1} \equiv a^{\phi(n)-1} \pmod{n}$ (co-prime a)

WORKED EXAMPLE : exponent reduction by Euler

Q. Using Euler's theorem, find (a) $11^{2024} \bmod 35$ (b) the last two digits of 3^{1003} (c) $20^{62} \bmod 77$ (slide example).
A.

	gcd, $\phi(n)$	Exponent	Remaining power	Result
(a)	$\gcd(11,35)=1$, $\phi(35)=4\cdot6=24$	$2024=84\cdot24+8$	$11^2=121\equiv16$, $11^4\equiv16^2=256\equiv11$, $11^8\equiv11^2\equiv16$	**16**
(b)	$\gcd(3,100)=1$, $\phi(100)=40$	$1003=25\cdot40+3$	$3^3=27$	**27**
(c)	$\gcd(20,77)=1$, $\phi(77)=6\cdot10=60$	$62=1\cdot60+2$	$20^2=400=5\cdot77+15$	**15**

For (b), "last two digits" means the residue modulo 100. **Check** (by script): $11^{2024} \bmod 35 = 16$, $3^{1003} \bmod 100 = 27$, $20^{62} \bmod 77 = 15$.

1.3 Multiplicative inverses via Fermat and Euler ★

★ **In the papers:** 2024–25 Q5(iii): $7^{-1} \bmod 17$, $15^{-1} \bmod 99$ (does not exist).

The multiplicative inverse of a modulo n is the residue a^{-1} with $a \cdot a^{-1} \equiv 1 \pmod{n}$; it exists if and only if $\gcd(a,n)=1$ (derived in §**??**). It depends on the modulus as well as on a: 2^{-1} is 3 modulo 5 but 4 modulo 7. For small n it can be found by trial, multiplying a by $1, 2, 3, \dots$ until the product is $\equiv 1$:

a mod n	gcd	Search	Inverse	Note
3 mod 5	1	$3 \cdot 2 = 6 \equiv 1$	**2**	so $2^{-1} \bmod 5 = 3$ as well
2 mod 11	1	$2 \cdot 6 = 12 \equiv 1$	**6**	
4 mod 5	1	$4 \cdot 4 = 16 \equiv 1$	**4**	its own inverse ($4 \equiv -1$)
2 mod 7	1	$2 \cdot 4 = 8 \equiv 1$	**4**	not 3: depends on n
5 mod 10	**5**	$5k \bmod 10 \in \{0, 5\}$ only	**none**	

Inverse by exponentiation. Fermat's theorem already contains the inverse. If p is prime and $p \nmid a$, then $a \cdot a^{p-2} = a^{p-1} \equiv 1$, so

$$a^{-1} \equiv a^{p-2} \pmod{p}, \qquad \text{and in the same way} \qquad a^{-1} \equiv a^{\phi(n)-1} \pmod{n} \text{ if } \gcd(a, n) = 1.$$

This replaces the extended Euclidean algorithm by one modular exponentiation, which is convenient when the modulus is prime or when $\phi(n)$ is known. The price is that for a composite n the factorisation is needed to get $\phi(n)$. The slides' examples (L10-15 and L10-20), all checked by script:

Fermat (n prime)		Euler (n composite)		
$8^{-1} \bmod 17 = 8^{15}$	$= 15$	$8^{-1} \bmod 77 = 8^{59}$	$= 29$	$\phi(77) = 60$
$5^{-1} \bmod 23 = 5^{21}$	$= 14$	$7^{-1} \bmod 15 = 7^{7}$	$= 13$	$\phi(15) = 8$
$60^{-1} \bmod 101 = 60^{99}$	$= 32$	$60^{-1} \bmod 187 = 60^{159}$	$= 53$	$\phi(187) = 160$
$22^{-1} \bmod 211 = 22^{209}$	$= 48$	$71^{-1} \bmod 100 = 71^{39}$	$= 31$	$\phi(100) = 40$
$2^{-1} \bmod 13 = 2^{11}$	$= 7$	$3^{-1} \bmod 20 = 3^{7}$	$= 7$	$\phi(20) = 8$

Each answer is checked by one multiplication, e.g. $8 \cdot 15 = 120 = 7 \cdot 17 + 1$ and $7 \cdot 13 = 91 = 6 \cdot 15 + 1$.

Trap: the inverse may not exist

Before using any formula, compute $\gcd(a, n)$. If it is not 1 there is no inverse, and the exam expects exactly that answer with the reason. Blindly applying Euler's formula still produces a number, which is wrong: for $15^{-1} \bmod 99$, $\phi(99) = 60$ and $15^{59} \bmod 99 = 36$, but $15 \cdot 36 = 540 \equiv 45 \not\equiv 1 \pmod{99}$.

WORKED EXAMPLE : 2024–25 Q5(ii) and (iii), as it should be written

Q. (ii) Find the values of $100^{95} \bmod 9$, $658^{17} \bmod 17$ and $145^{102} \bmod 101$. (iii) Find the values of $7^{-1} \bmod 17$ and $15^{-1} \bmod 99$.
A. (ii) $100^{95} \bmod 9$. The modulus 9 is not prime, so Fermat's theorem cannot be used; reduce the base instead. $100 = 11 \cdot 9 + 1$, so $100 \equiv 1 \pmod 9$ and $100^{95} \equiv 1^{95} = \mathbf{1}$. (Euler agrees: $\gcd(100, 9) = 1$, $\phi(9) = 6$, $95 \bmod 6 = 5$, $1^5 = 1$.)
$658^{17} \bmod 17$. 17 is prime, so by the second version of Fermat's theorem $a^{17} \equiv a \pmod{17}$. $658 = 38 \cdot 17 + 12$, hence $658^{17} \equiv 658 \equiv \mathbf{12}$.
$145^{102} \bmod 101$. 101 is prime and $101 \nmid 145$. Base: $145 \equiv 44$. Exponent: $102 = 1 \cdot 100 + 2$, and $44^{100} \equiv 1$ by Fermat. So $145^{102} \equiv 44^2 = 1936 = 19 \cdot 101 + 17$, i.e. **17**.
(iii) $7^{-1} \bmod 17$. 17 is prime and $17 \nmid 7$, so $7^{-1} \equiv 7^{17-2} = 7^{15} \pmod{17}$. With $15 = 8 + 4 + 2 + 1$:

7^1	7^2	7^4	7^8	$7^{15} = 7^8 \cdot 7^4 \cdot 7^2 \cdot 7$
7	$49 \equiv 15$	$15^2 = 225 \equiv 4$	$4^2 = 16$	$16 \cdot 4 = 64 \equiv 13$; $13 \cdot 15 = 195 \equiv 8$; $8 \cdot 7 = 56 \equiv \mathbf{5}$

So $7^{-1} \bmod 17 = \mathbf{5}$. Check: $7 \cdot 5 = 35 = 2 \cdot 17 + 1$ ✓.
$15^{-1} \bmod 99$. $\gcd(99, 15)$: $99 = 6 \cdot 15 + 9$, $15 = 1 \cdot 9 + 6$, $9 = 1 \cdot 6 + 3$, $6 = 2 \cdot 3 + 0$, so $\gcd = 3 \neq 1$. Every number $15x - 99k$ is a multiple of 3 and can never equal 1, therefore $15^{-1} \bmod 99$ **does not exist**.

Which method? All four methods give the same inverse when it exists; they differ in what they need. The summary below is the decision to make in an exam.

Method	Formula	When to use
Trial	$a \cdot 1, a \cdot 2, \ldots$ until $\equiv 1$	Small n only (mod 26 ciphers, GF(7) division).
Fermat	$a^{-1} = a^{p-2} \bmod p$	Prime modulus; compute with square-and-multiply.
Euler	$a^{-1} = a^{\phi(n)-1} \bmod n$	Composite n whose factorisation (hence $\phi(n)$) is known.
Extended Euclid	$sa + tn = 1 \Rightarrow a^{-1} = s \bmod n$	The general method: works for any n without factoring, and detects "no inverse" (final gcd $\neq 1$). The past papers ask for it explicitly (2024–25 Q7, §**??**); RSA uses it to find d.

1.4 Order of an element ▸

▸ **Likely this year:** "Find the order of every element of $\mathbb{Z}_7^*$ / $\mathbb{Z}_{10}^*$" (Forouzan), usually as the first step of a primitive-root question.

Euler's theorem says that $a^{\phi(n)} \equiv 1$ for every $a \in \mathbb{Z}_n^*$, but many elements return to 1 much earlier. The order of a modulo n, $\text{ord}(a)$, is the smallest positive integer i with $a^i \equiv 1 \pmod{n}$. It exists for every $a \in \mathbb{Z}_n^*$, and Euler's theorem guarantees $\text{ord}(a) \le \phi(n)$. After $a^{\text{ord}(a)} = 1$ the powers start over, so the sequence $a, a^2, a^3, \ldots$ is periodic with period exactly $\text{ord}(a)$.

The key property: $a^m \equiv 1 \iff \text{ord}(a) \mid m$. Write $m = q \cdot \text{ord}(a) + r$ with $0 \le r < \text{ord}(a)$. Then $a^m = \left(a^{\text{ord}(a)}\right)^q a^r \equiv a^r$, and by the minimality of the order $a^r \equiv 1$ only for $r = 0$. Two consequences follow. First, since $a^{\phi(n)} \equiv 1$, **the order always divides** $\phi(n)$ (this is Lagrange's theorem for the group $\mathbb{Z}_n^*$), so to find an order only the divisors of $\phi(n)$ need to be tried. Second, exponents can be reduced modulo $\text{ord}(a)$, which is sharper than reducing modulo $\phi(n)$: $\text{ord}(2) = 3$ modulo 7, so $2^{100} \equiv 2^{100 \bmod 3} = 2^1 = 2 \pmod{7}$.

The table lists $a^i \bmod 7$ for the whole group $\mathbb{Z}_7^*$ (slide L11-30); the shaded cell is the first 1 in each row, and its column is the order. Every order is a divisor of $\phi(7) = 6$, and only $a = 3$ and $a = 5$ reach the maximum 6.

a	$i=1$	2	3	4	5	6	$\text{ord}(a)$
1	**1**	1	1	1	1	1	1
2	2	4	**1**	2	4	1	3
3	3	2	6	4	5	**1**	**6**
4	4	2	**1**	4	2	1	3
5	5	4	6	2	3	**1**	**6**
6	6	**1**	6	1	6	1	2

Two more slide examples (L11-27, L11-29): in $\mathbb{Z}_{10}^* = \{1,3,7,9\}$ the orders are $1, 4, 4, 2$ ($3^2 = 9$, $3^4 = 81 \equiv 1$), and in $\mathbb{Z}_8^* = \{1,3,5,7\}$ they are $1, 2, 2, 2$, because $3^2 = 9$, $5^2 = 25$ and $7^2 = 49$ are all $\equiv 1 \pmod{8}$.

1.5 Primitive roots ▸

▸ **Likely this year:** "Find all primitive roots of 7 / 11 / 13", "is 3 a primitive root of 17?", "how many primitive roots does 17 have?" (needed for Diffie–Hellman and ElGamal).

The idea, for a prime p**.** Take the multiplicative group $\mathbb{Z}_p^* = \{1, 2, \ldots, p-1\}$, which is $\mathbb{Z}_p$ without 0. Pick an element g and write down its powers $g^1, g^2, g^3, \ldots \pmod{p}$. Two things can happen. Either the powers run through *every* element of $\mathbb{Z}_p^*$ before returning to 1, or they hit 1 early and then repeat a smaller cycle. In the first case g is called a primitive root of p: its powers $g^1, \ldots, g^{p-1}$ produce each of $1, \ldots, p-1$ exactly once. Modulo 7, for example:

	g^1	g^2	g^3	g^4	g^5	g^6	
$g = 3$	3	2	6	4	5	1	all six elements: **primitive root**
$g = 2$	2	4	1	2	4	1	only $\{2,4,1\}$: **not** a primitive root

Since the powers start repeating as soon as they reach 1, the number of different powers equals the order of g. Producing all $p-1$ elements therefore means exactly that $\text{ord}(g) = p-1$, the largest order possible. When such a g exists, every element of the group is a power of it, so the group is *cyclic* and g is a *generator*. Every element y is then g^x for exactly one $x \in \{1, \ldots, p-1\}$, and that x is the discrete logarithm of y (§**??**).

Why cryptography needs them. Diffie–Hellman and ElGamal publish p and a base g, keep an exponent x secret, and publish $g^x \bmod p$. If g had a small order d, then g^x could take only d different values, and Eve could find a matching exponent by trying d values. Choosing a primitive root makes all $p-1$ values possible, so the search is as large as it can be.

The same idea for any n**.** Nothing changes except the group. For a general n the group is $\mathbb{Z}_n^*$, the numbers below n that are coprime to n, and it has $\phi(n)$ elements instead of $p-1$. So $g \in \mathbb{Z}_n^*$ is a primitive root of n if $\text{ord}(g) = \phi(n)$, which means $g^1, \ldots, g^{\phi(n)}$ produce every element of $\mathbb{Z}_n^*$ once each. Modulo 10, $\mathbb{Z}_{10}^* = \{1,3,7,9\}$ and the powers of 3 are $3, 9, 7, 1$, so 3 is a primitive root of 10. The powers never produce 0, 2, 4, 5, 6 or 8, and they are not supposed to: a power of a number coprime to n is itself coprime to n, so only $\mathbb{Z}_n^*$ can ever be generated, never all of $\mathbb{Z}_n$. For a prime this is the same statement as before, since $\mathbb{Z}_p^*$ is $\mathbb{Z}_p$ without 0 and $\phi(p) = p-1$.

Unlike primes, many composite n have *no* primitive root at all, because every element returns to 1 too early. Modulo 8, the group $\mathbb{Z}_8^* = \{1,3,5,7\}$ has $\phi(8) = 4$ elements, but 3^2, 5^2 and 7^2 are all $\equiv 1$, so no order reaches 4 (slide L11-28). Modulo 15, every element satisfies $a^4 \equiv 1$, because $a^2 \equiv 1 \pmod{3}$ and $a^4 \equiv 1 \pmod{5}$, so no order reaches $\phi(15) = 8$. The complete answer is a theorem worth memorising:

★ MUST KNOW : primitive roots

g is a primitive root of $n \iff \text{ord}(g) = \phi(n) \iff$ the powers of g produce all of $\mathbb{Z}_n^*$ (for a prime: all of $1, \ldots, p-1$). Primitive roots exist **only** for $n = 2,\ 4,\ p^k,\ 2p^k$ (p an odd prime). Every prime has them; 9, 10, 18 have them; 8, 12, 15 do not.

When they exist there are exactly $\phi(\phi(n))$ of them; for a prime p that is $\phi(p-1)$.

Test 1: list the powers. Multiply by g repeatedly, reducing each time; for example $3^4 \equiv 6 \cdot 3 = 18 \equiv 4 \pmod 7$, so you never need $3^6 = 729$. If a 1 appears before the $(p-1)$-th power, stop: g is not a primitive root.

Test	Powers $g^1, g^2, \ldots \pmod p$	Verdict
2 mod 5	2, 4, 3, 1	order $4 = p-1$: **yes**
3 mod 5	3, 4, 2, 1	**yes**
4 mod 5	4, **1**	order 2: no
3 mod 7	3, 2, 6, 4, 5, 1	**yes**
2 mod 7	2, 4, **1**	order 3: no
2 mod 11 (homework)	2, 4, 8, 5, 10, 9, 7, 3, 6, 1	**yes**

The homework "primitive roots of 5" is answered by the first three rows: 1 never works ($1^k = 1$), 2 and 3 do, and 4 does not, so the primitive roots of 5 are **2 and 3**.

Test 2: the fast test. Listing takes up to $p-1$ multiplications. It is faster to use the fact that the order of g must divide $p-1$: if g is not a primitive root, its order is a proper divisor of $p-1$, and every proper divisor divides $(p-1)/q$ for some prime factor q of $p-1$. So it is enough to check those few exponents:

$$g \text{ is a primitive root of } p \iff g^{(p-1)/q} \not\equiv 1 \pmod p \text{ for every prime } q \text{ dividing } p-1.$$

For $p = 7$, $p-1 = 6 = 2 \cdot 3$, so check g^3 and g^2. For $g = 3$: $3^3 = 27 \equiv 6$ and $3^2 = 9 \equiv 2$, neither is 1, so 3 is a primitive root. For $g = 2$: $2^3 = 8 \equiv 1$, so 2 is not.

Finding all of them from one. Once one primitive root g is known, the others are the powers g^k with $\gcd(k, p-1) = 1$. The reason is that g^k has order $\frac{p-1}{\gcd(k,p-1)}$, which equals $p-1$ exactly when the gcd is 1. Counting those k gives the $\phi(p-1)$ from the box: 5 has $\phi(4) = 2$ (2, 3), 7 has $\phi(6) = 2$ (3, 5), 11 has $\phi(10) = 4$ (2, 6, 7, 8), and 13 has $\phi(12) = 4$ (2, 6, 7, 11).

WORKED EXAMPLE : all primitive roots of a prime

Q. Find all the primitive roots of 11 (i.e. of the group $G = \langle \mathbb{Z}_{11}^*, \times \rangle$).
A. $\phi(11) = 10 = 2 \cdot 5$, so a primitive root g must satisfy $g^{10/2} = g^5 \not\equiv 1$ and $g^{10/5} = g^2 \not\equiv 1$. Try $g = 2$: $2^2 = 4 \not\equiv 1$ and $2^5 = 32 = 2 \cdot 11 + 10 \equiv 10 \not\equiv 1$, so **2 is a primitive root**. (For contrast, $g = 3$ fails: $3^5 = 243 = 22 \cdot 11 + 1 \equiv 1$.)
The other primitive roots are 2^k with $\gcd(k, 10) = 1$, i.e. $k \in \{1, 3, 7, 9\}$:

k	1	3	7	9
$2^k \bmod 11$	2	8	$128 = 11 \cdot 11 + 7 \to 7$	$512 = 46 \cdot 11 + 6 \to 6$

The primitive roots of 11 are **2, 6, 7, 8**. **Check:** their number is $\phi(10) = 4$ ✓, and the full list of powers of 2 is $2, 4, 8, 5, 10, 9, 7, 3, 6, 1$, all ten residues.

WORKED EXAMPLE : fast test and counting

Q. Show that 3 is a primitive root of 17 but 2 is not. How many primitive roots does 17 have? List them.
A. $\phi(17) = 16 = 2^4$, whose only prime factor is 2, so g is a primitive root iff $g^{16/2} = g^8 \not\equiv 1 \pmod{17}$. For $g = 3$: $3^2 = 9$, $3^4 = 81 \equiv 13$, $3^8 \equiv 13^2 = 169 \equiv 16 \not\equiv 1$, so **3 is a primitive root**. For $g = 2$: $2^4 = 16$, $2^8 \equiv 16^2 = 256 = 15 \cdot 17 + 1 \equiv 1$, so **2 is not** (its order is 8). The number of primitive roots is $\phi(16) = \mathbf{8}$. They are 3^k for odd k: $3^1, 3^3, \ldots, 3^{15} \equiv 3, 10, 5, 11, 14, 7, 12, 6$, i.e. **3, 5, 6, 7, 10, 11, 12, 14**.

Corrections to the notes / slides

- All values recomputed by script. The slides' inverse examples (L10-15: 15, 14, 32, 48; L10-20: 29, 13, 53, 31) and examples $6^{10} \bmod 11 = 1$, $3^{12} \bmod 11 = 9$, $6^{24} \bmod 35 = 1$, $20^{62} \bmod 77 = 15$ are correct. The video homework answers are correct: $5^{10} \equiv 1 \pmod{11}$, $11^{20} \equiv 22 \pmod{33}$, 2 is a primitive root of 11, the primitive roots of 5 are 2 and 3.
- Slide L10-16 says the second version of Euler's theorem "removes the condition that a and n should be coprime". That is true only for squarefree n (such as $n = pq$ in RSA). It fails for $n = 4$, $a = 2$: $2^{\phi(4)+1} = 2^3 = 8 \equiv 0 \neq 2$.
- Slide L10-18 solves $20^{62} \bmod 77$ with the second version, but $\gcd(20, 77) = 1$, so the first version already gives $20^{62} = 20^{60} \cdot 20^2 \equiv 400 \equiv 15$.
- Video notation: "-2^{12}" and "-1^{10}" mean $(-2)^{12}$ and $(-1)^{10}$. Without brackets $-2^{12} = -(2^{12})$; write the brackets in the exam.
- The general definition of a primitive root (any n) was stated in the video but not explained: only $n = 2, 4, p^k, 2p^k$ have primitive roots, and then there are $\phi(\phi(n))$ of them.

Check your understanding

1. **State both versions of Fermat's little theorem and the difference between them.** If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$; for every a, $a^p \equiv a \pmod{p}$. The second version drops the condition $p \nmid a$, since for $p \mid a$ both sides are 0.
2. **Why is the exponent reduced modulo $p - 1$ and not modulo p?** Because the powers of a repeat when the exponent grows by $p - 1$ ($a^{p-1} \equiv 1$), not by p. For example $2^{2026} \bmod 11$: $2026 \bmod 10 = 6$ and $2^6 = 64 \equiv$ **9**.
3. **Differentiate Fermat's and Euler's theorems.** Fermat needs a prime modulus and gives period $p - 1$; Euler allows any modulus n with $\gcd(a, n) = 1$ and gives period $\phi(n)$. Fermat is the special case $n = p$ of Euler.
4. **Find $3^{-1} \bmod 20$ by Euler's theorem.** $\gcd(3, 20) = 1$, $\phi(20) = 8$, so $3^{-1} \equiv 3^7 = 2187 \equiv$ **7**; check $3 \cdot 7 = 21 \equiv 1$.
5. **Why does $15^{-1} \bmod 99$ not exist, and what goes wrong if Euler's formula is used anyway?** $\gcd(15, 99) = 3$, so $15x - 99k$ is always a multiple of 3. The formula gives $15^{59} \bmod 99 = 36$, but $15 \cdot 36 \equiv 45 \neq 1$.
6. **Why does RSA decryption recover the message?** $ed = k\phi(n) + 1$, so $(M^e)^d = M^{k\phi(n)+1} \equiv M \pmod{n}$ by the second version of Euler's theorem, which holds for $n = pq$ even if $\gcd(M, n) \neq 1$.
7. **What is the order of 2 modulo 13, and is 2 a primitive root?** The order divides 12; $2^4 \equiv 3$ and $2^6 = 64 \equiv 12$, so neither $12/3 = 4$ nor $12/2 = 6$ gives 1. Hence $\mathrm{ord}(2) = 12$ and 2 is a primitive root of 13. The others are $2^5 \equiv 6$, $2^7 \equiv 11$, $2^{11} \equiv 7$, so the primitive roots of 13 are 2, 6, 7, 11.
8. **How many primitive roots does 23 have, and is 5 one of them?** $\phi(22) = 10$. Since $22 = 2 \cdot 11$, check $5^{11} \equiv 22 \neq 1$ and $5^2 = 25 \equiv 2 \neq 1$, so 5 is a primitive root.
9. **Why does $\mathbb{Z}_8^*$ have no primitive root?** $\phi(8) = 4$, but 3^2, 5^2 and 7^2 are all $\equiv 1 \pmod{8}$, so no element has order 4.
10. **Why must the Diffie–Hellman base be a primitive root?** If g had order $d < p - 1$, then g^x would take only d values and the secret exponent could be found among d candidates; a primitive root makes g^x range over all $p - 1$ residues.